

SCOPE AND DEFINITIONS:

Read this manual before you operate the Vacuum Coating unit & its accessories. Important safety information is highlighted as **WARNING** and **CAUTION** instructions. You must obey these instructions. The uses of **WARNING** and **CAUTION** are defined below.

WARNING

Warnings are given where failure to observe the instruction could result in injury or death to people.

CAUTION

CAUTIONS are given where failure to observe the instruction could result in damage to the equipment, associated equipment and process.

OPERATOR SAFETY:

WARNING

Obey the safety instructions given below and take note of appropriate precautions. If you do not, you can cause injury to people and damage to equipment.

- Read all of the relevant instructions before you operate any accessories.
- Surfaces within the system may be very hot or very cold. Do not touch hot or cold surfaces such as the source, substrate holder, diffusion pump, liquid nitrogen trap, Rotary Pump etc.
- Observe all safety precautions when you come into contact with dangerous substances, which have been used with the source.
- Wear clean lint-free gloves when you handle components in the chamber to prevent the contamination of the source and its accessories.
- Do not operate the source in an area where there is equipment subject to magnetic interference.

1.0 INTRODUCTION:

HINDHIVAC Vacuum coater model **BC-300** can perform a large number of industrial and laboratory applications like preparation of thin films for optical and electronic applications, preparation of specimens for Electron Microscope, etc. The basic unit consists of cabinet containing a vacuum pumping system together with all the electrical components necessary for the coating process.

2.0 SYSTEM DESCRIPTION

2.1 VACUUM CHAMBER

Vacuum chamber is a box type and is made of non-magnetic quality AISI-304 stainless steel processed for very high vacuum.

- a. Nominal internal size: 395 (w) x 360 mm (d)
- b. Nominal external size: 400 (w) x 400 mm (d)
- c. Nominal height of the chamber 350mm External
- d. Nominal height of the chamber 300mm (internal)

Top plate is provided with six blanked off 25mm bore holes suitable for fitting accessories such as rotary work holder, quartz crystal monitor sensor, radiant heater feed through, thermocouple feed through.

One No. of 2" size view port with toughened glass is fitted on the Top Plate in addition to the view port on the door.

Bottom of the chamber is welded with base plate having 11 Nos. of feed through for fixing the mechanical and electrical feed through and the same is mounted above the vacuum collar of the pumping system.

Five piece set, removable stainless steel liners are provided for easy system cleaning with minimum down time.

2.2 PUMPING SYSTEM

The chamber is evacuated by a **HINDHIVAC** diffusion pump model OD-150D (having pumping speed of 750 lps) and backed by double stage, direct driven, rotary vacuum pump, Model ED-21.

THE SYSTEM CONTAINS FOLLOWING VACUUM VALVES:

- a. A hand operated high vacuum valve is fixed to a base plate of 13" dia. This valve isolates the chamber from the pumping system so that the chamber can be brought to atmospheric pressure without switching off the pumping system. A stainless steel wire mesh is fixed over the base plate opening to prevent foreign bodies from falling into the high vacuum valve.
- b. A single handle operated 3 way combination valve is used in roughing and backing line. This combined roughing and backing valves are interlocked due to single lever operation they cannot be opened simultaneously.
- c. An air admittance valve is fixed to the collar pipeline to release the chamber vacuum after each coating process cycle.
- d. A fine control needle valve is also provided on the collar for admitting process gas during H.T (Ion Bombardment) cleaning of substrate.

All the above mentioned valves are positioned on the front panel for Operation.

3.0 SAFETY DEVICES & DESCRIPTION

- a. An over load relay for over current protection for rotary vacuum pump motor is provided.
- b. The cooling coil of the diffusion pump is equipped with a thermal switch, which isolates the power supply to the pump heater in the event of over heating of pump.
- c. A water flow switch is incorporated in the outlet of the cooling water supply line to switch "OFF" the power to the diffusion pump in case of failure of water supply or minimum recommended flow pressure.
- d. A vacuum switch is incorporated in the roughing line & set to operate when the chamber pressure approaches atmospheric pressure & isolate the power supplies (HT,LT,RH,SH). This prevents accidental "ON" of HT power supply when the chamber is exposed to atmosphere.

- e. The LT supply for evaporation filaments or boats is obtained from 230V input transformer by means of parallel or series connections in the secondary side of the transformer. The output rating may be either 20V 100Amps intermittent or 10V at 200Amps. The standard unit is normally connected for 200 amps rating. The LT output from the transformer is fed through a current meter and through a selector switch to LT feed through and filament holders.
- f. The doors of the coating unit cabinet are equipped with a switch to isolate the power supply of HT, RH & LT when door is removed (refer circuit diagram enclosed).
- g. The HT supply for glow discharge cleaning (ion bombardment) is obtained from a high reactance transformer rated at 3.5 KV 50ma (5KV AC open circuit).
- h. The input voltage to the LT and HT transformers and first selected with a switch on the electrical control panel and then fed to a manually operated HT CONTROL knob on the front panel coating unit. A circuit breaker in the return line is provided to protect the HT control from overload. A 0 to 10 amps AC primary current ammeter is also provided in the return line to measure the primary current.
- i. To measure evaporation current (LT secondary) an ammeter is provided on the panel. It works in conjunction with a current transformer, which is connected to the secondary of the LT transformer.

FUSES:

FH1	2A	MAINS
FH2	6.3A	Control transformer
FH3	6.3A	Control circuit

VACUUM MEASUREMENT:

A Pirani operates in conjunction with two Pirani gauge heads. One mounted in the rotary vacuum pump pipeline and the other in the collar side. The Pirani gauge

measures from 0.5m bar to 1×10^{-3} m bar and the Pirani gauge is used to measure roughing and backing pressure.

The penning gauge head is mounted in the collar. The penning gauge head measures from 1×10^{-3} m bar to 1×10^{-6} m bar. The penning gauge is used to measure high vacuum in the chamber.

SIX TELEMECHINIC SELECTOR SWITCHES ARE LABELLED AS UNDER

- 1. ROTARY PUMP (To switch either ON or OFF)**
- 2. DIFFUSION PUMP (To switch either ON or OFF)**
- 3. HT CONTROL (To switch either ON or OFF)**
- 4. LT CONTROL (To switch either ON or OFF)**

CHAMBER GADGETORY (STANDARD)

- A. The chamber gadgetry comprises of a work holder ring which has a diameter of 320 mm. The work holder ring is supported by rotary drive shaft fixed to the top plate. A spherical work holder which is supplied unpaired along with the unit is mounted on the work holder ring for uniform coatings on plain area from a central evaporation source.
- B. A DC high-tension discharge cleaning system consisting of pure aluminum bar (cathode) with suitable shielded to avoid electron contamination of the substrate.
- C. A source shutter swinging over the source position and operated by an external lever is provided on the top of the tabletop.
- D. A standard filament holder is fixed to the LT live electrode and an earth electrode. The filament is normally positioned vertically below the center of the work holder to give uniform distribution of the evaporation. Two sets of off-Sector filament holders and one set of center filament holder are supplied fix them these three sets of LT electrodes for multi layer coatings on small plane areas where there is no uniformity. A 200amps selector switch is provided on the door to select any one of the three LT electrodes.
- E. A baffle plate is provided within the tripod just above the base plate aperture. This also acts as a radiation shield when the radiant heater is used.

- F. A stainless steel wire mesh is provided over the base plate opening to prevent foreign bodies from falling into the baffle valve.

4.0 ACCESSORIES (OPTIONAL)

4.1 ION BOMBARDMENT

This high-tension discharge cleaning system consists of a super pure aluminum electrode bar. The bar is shielded to avoid electron contamination of the substrate during discharge cleaning.

4.2 SOURCE SHUTTER PLATE

This plate is designed to cover or uncover the off sector filament holders meant for sequential evaporation when a rotary drive is used. The shutter plate is attached to a standard source shutter shaft when a rotary drive is used.

4.3 MANUAL OPERATED SOURCE SHUTTER

The manually operated source shutter is used to cover or uncover the evaporation source which mounted on top of stainless steel pillars fitted to the base plate for support is provided in the chamber.

UTILITIES REQUIRED

Power supply	: 230V AC single phase 50 Hz 15 amps (see circuit Diagram) Maximum power consumption about 3 KVA.
Water at 20°C	: Consumption 2 lpm for DP, DTM & Metal Chamber.
Liquid Nitrogen	: 5 liters for 8 hours operation as required by customer
Installation	: The unit should be installed on rigid and even level concrete floor.
Unpacking	: The unit, in a dismantled condition, is packed carefully in several boxes with chemical foams to avoid damage during transportation. On receipt of the unit, carefully unpack and check the Components for damage.

ROTARY VACUUM PUMP

Before connecting the rotary vacuum pump to the coating unit, charge the oil to the pumps as Follows.

- A. Remove the oil pouring dummy of the rotary pump.
- B. Fill the pump with oil to the level marked on the sight glass window in the pump wall, and replace the oil pouring dummy.

DIFFUSION PUMP

- a). Charge the diffusion pump with 200cc of the silicon oil supplied as follows.
- b). Vent Air into the diffusion pump.
- c). Before pouring the oil pre-heat the Diffusion pump for at least 10 minutes. And cool it about half an hour.
- d). Remove the knurled lock nut on the oil drain/pour port of the diffusion pump.
- e). Slowly pour the 200 cc. of diffusion pump oil into the oil port using a clean funnel
- f). Re-position the dummy and tighten the joints carefully. (Make sure that you have placed the 'O' ring properly).

Note: The vacuum pipe lines and the base plate configurations have to be assembled by referring the photographs provided in this manual

5.0 INITIAL SETTING UP

Prior to starting a newly installed coating unit, carry out the following.

- a) Check the vacuum gauges in accordance with the relevant working instructions supplied. The operator should be thoroughly familiar with it before using the coating unit.
- b) Fix new filaments or evaporation boats to the filament holders. Whenever a new filament or boat is fixed the chamber must first be

evacuated to 10^{-5} m.bar and flushed the same to remove contamination etc. by passing slightly higher current than that is normally used for evaporation.

- c) The rotary work holder if fitted can be operated by means of the rotary switch (on the electrical panel) and speed control.
- d) Check whether the rotary vacuum pump oil level is correct.
- e) Check all the manually operated valves are closed.
- f) Check all the switches on the control panel are 'OFF'
- g) Check all the coating unit cabinet doors are properly closed.

6.0 OPERATION

1. Switch 'ON' the main switch, the "Mains On" indication lamp should glow.
2. Switch 'ON' the rotary vacuum pump by selector switch, "RP" indication lamp should glow.
3. Switch 'ON' the Pirani gauge.
4. On the first start of each day, check the rotary pump vacuum by means of gauge head-1 in Pirani gauge.
5. Turn the combination valve to backing position. The pressure should fall steadily to better than 0.05 m.bar as the diffusion pump is evacuated.
6. Allow the cooling water supply to flow to DP and chamber.
7. Switch 'ON' the Diffusion pump by selector switch. The diffusion pump will take about 30 minutes to reach the operating temperature.
8. Admit NITROGEN GAS into the chamber and open the chamber.
9. Load the filaments or boats with the required material and fix the substrates on the work holder. Note down the materials that are loaded in the turret with the appropriate number of that particular filament or boat. Select the required filament or boat on turret (if turret is fitted).

10. Cover the filament or boat with source shutter.
11. Close the chamber.

NOTE: Clean the 'O' ring sealing portion before placing the chamber

12. After Diffusion pump is ready, close the chamber air admittance valve and needle valve.
13. Turn the combination valve lever to "Roughing" position.
14. Select gauge head-2 in Pirani.
15. Wait until gauge head-2 shows vacuum better than 0.05m.bar.
16. Turn the combination valve lever to "Backing" position.
17. Select gauge head-1 position in Pirani gauge.
18. Slowly open the high vacuum valve so that High vacuum is created in the chamber.
19. Switch ON the Penning gauge to read high vacuum in the chamber.
20. Once the required vacuum is reached proceed with your coating process/IB cleaning process etc.

7.0 NOTES ON HT CLEANING

Adhesive qualities and durability of the deposited films can be greatly improved by thoroughly cleaning the substrate surface. For glass, after initial cleaning by detergents and solvents, the surface must be freed from absorbed gases by a high-tension electrical discharge.

The glow discharge also removes any absorbed vapors and gases from the chamber's internal surface with a consequent reduction in pump down time.

If the work is liable to damage by heat the discharge must be used carefully with lower intensity so that the bombardment is not too intense.

In order to maintain high-tension discharge in the chamber, the pressure must not be below a minimum of 0.05 mbar. The minimum pressure is determined by the applied voltage and the distance between HT electrodes. In order to maintain a pressure in the chamber above this minimum while continuing to pump away gases and vapors the pumping speed must be throttled and a small constant flow of air admitted to the chamber through a needle valve. Throttling is achieved by opening the baffle valve partially. Fine adjustments of pressure in the chamber during HT cleaning are then carried out by the needle valve opening or closing. If HT discharge cleaning is to be used, it should be carried out at this state as follows.

NOTE: Prior to HT cleaning, select DP and RD position. Switch 'ON' the rotary drive speed control. Leave the work holder rotating throughout the cleaning and evaporation cycles (if rotary drive is fitted).

21. Select gauge head -2 Pirani gauge.
22. Switch 'OFF' the penning gauge.
23. Open the needle valve slowly and maintain pressure in gauge head-2 between 0.05 m.bar to 0.1m.bar (High vacuum valve should be in partially open).
24. Switch ON HT by selecting HT rotary switch to HT position. Switch ON CB1.
25. Increase the "HT/LT CONTROL" slowly. Measure the primary current of HT on the primary current meter. Set the current at about 1 Amp for the required cleaning time.
26. Do not allow the backing pressure, i.e. in gauge head-1 to exceed 0.1m.bar at any time.
27. Using the needle valve to control the pressure in the chamber.
28. Glow discharge time depends upon the type of substrate. For normal application 5 to 7 minutes is sufficient.

29. On completion of HT cleaning bring the "HT CONTROL" to zero position, switch OFF HT and CB-1.
30. Close the needle valve and open high vacuum valve slowly keeping the backing pressure from rising above 0.1m.bar.
31. Switch 'ON' the penning gauge.
32. Select range-1.

NOTE: Sometimes the ionization phenomena by which the Penning gauge works will not initiate as soon as the gauge is switched 'ON'. It takes a few seconds to initiate the discharge. So keep it in 'ON' position in case it dose not initiate immediately. After the discharge initiates the meter needle shows the correct pressure in chamber.

33. When pressure drops in the chamber better than 10^{-5} m.bar., if radiant heater is to be used, switch 'ON' the radiant heater (RH) by selector switch RH 'ON'.
34. The temperature can be monitored with the help of K type thermocouple in conjunction with a digital PID controller directly in deg.C.
35. After attaining the required temperature the heater temp will be automatically controlled by digital PID controller. Switch off RH and when heating process is over.
36. Allow the system to pump down the chamber below 10^{-5} mbar. And select the filament holder by selecting the selector switch.
37. Switch 'ON' LT control switch and also switch 'ON' LT CB1.
38. Slowly turn the "LT CONTROL" knob (with source shutter covered) and observe the evaporation through the front window, when evaporate is just melting uncover the source shutter unless initial wetting of filament is needed in which case leave the source covered until this is complete.
39. Repeat the process with each filament if necessary. Minimize the "LT CONTROL" knob before changing the filament selector switch each time.

40. When evaporation is completed close the source shutter. Minimize the "LT CONTROL" knob. Switch off the LT and CB-1.
41. Switch off the penning gauge, close the high vacuum valve and allow the chamber temperature to come down to 50°C or less in case the chamber is at higher temperature.
42. Open the air admittance valves till airflow ceases and then remove the chamber.
43. Unload the work holder. Handle the fresh films with care.

8.0 RE-CYCLING

Re-load the work holder and vapor source when necessary and proceed as before from starting the unit from SLNo-9.

9.0 SHUTTING DOWN THE UNIT.

The operator is advised to leave the chamber under vacuum at all time when the unit is not in use in order to prevent exposure of the chamber interior to atmosphere.

1. Evacuate the chamber down to 10^{-4} m.bar then close the high vacuum valve.
2. Switch 'OFF' the penning gauge.
3. Switch 'OFF' the diffusion pump.
4. Run the rotary pump for a period of 20-25 minutes keeping the backing valve open.
5. Close backing valve and switch-'OFF' the rotary vacuum pump.
6. Close the cooling water supply.
7. Switch of mains switch.

NOTE: Before attempting to re-open the chamber do not forget admit air first.

10.0 LEAK DETECTION

1. SPURIOUS LEAKS

If the equipment has been dismantled and re-assembled a longer pump down time must be allowed to deal with out gassing from the walls of the system and to pump, similarly contamination of the system with vapor given off by dust, greases, etc., will prolong the pump down time and give the appearance of leakage. This can be avoided by paying strict attention to the pumping components and its cleanliness.

THE FOLLOWING ITEMS SHOULD THEREFORE BE CHECK:

The diffusion pump heater for open circuit.

The Quantity and condition of the oil in the diffusion pump.

The interior cleanliness of the chamber system. The state of chamber sealing gasket and sealing surface.

INDICATION OF LEAKS

A. LENS BLOOMING

After deposition the condition of the residual magnesium fluoride will give an indication of the chamber vacuum. A good vacuum will cause the residue to appear white or light gray. If it appears dark gray or black, air leakage should be suspected. The hardness of the film and leak or uniform color are further indications of poor vacuum.

B. EVAPORATED METAL FILMS

Here again lack of brightness and or hardness in the film is an indication of poor vacuum. Also under good vacuum conditions, the deposited metal, cast sharp shadows of objects on the work chamber since the molecules of metal vapor travels in straight paths. The mean free path of gas molecules at low

pressure being sufficiently long to make collusions between them and the deposited metal negligible.

C. CONDITIONS OF FILAMENT WIRE

Both molybdenum and tungsten filaments appears bright after heating in good vacuum. Should the filament appear contaminated with a brownish deposit, leakage should be suspected.

TROUBLE SHOOTING

Trouble	Cause	Remedy
Mains indication lamp not glowing	a. Open power card lead b. Open fuse c. Faulty in main MCB d. Faulty lamp e. Faulty mains plug	a. Check & Replace b. Check & replace c. Check & replace d. Check & replace e. Check & replace
Rotary pump not working	a. Faulty in MCB b. Fault in start/stop button c. Faulty wire d. Faulty pump	a. Check & replace b. Check & replace c. Check & replace d. Check & replace
HT/LT not working	a. Rotary pump 'OFF' b. Faulty in MCB c. Faulty selector switch d. Door are open e. Open HT Transformer f. Faulty vacuum switch	a. Switch 'ON' RP b. Check &replace c. Check &replace d. Close the doors properly e. Check & rectify f. Check & rectify
DP not working	a. Water pressure is low b. Open water flow switch c. Faulty in MCB d. Open thermostat e. Loose connection in 'DP' f. Faulty in selector switch g. Open heater	a. Check water pressure b. Check &replace c. Check &replace d. Check &replace e. Tighten connect proper f. check & replace g. check & replace

Rotary pump vacuum not achieving	a. Leakage Vacuum line b. Low level Rotary pump Oil c. Contamination of rotary pump Oil	a. Check leakage & rectify b. Check oil level c. Check & replace
Backing vacuum not achieving	a. leakage Vacuum line b. GH-1,CV-25 'O' rings sealing leak	a. Check leakage & rectify b. Check & replace
High vacuum not achieving	a. DP oil contamination b. Leakage in base plate assy c. Leakage in needle valve pipe line d. Leakage in air admittance vacuum line valve e. Leakage in belljar f. Sealing in 'L' gasket g. Belljar view port sealing leak	a. Replace oil b. Check leakage in base plate assy c. Check sealing in needle valve d. Check sealing in air admittance valve e. Check belljar leakage f. Clean and replace g. Check & replace